

Healthcare Data Integrity & Operational Reliability

Transforming raw transaction logs into actionable clinical insights to enhance hospital efficiency and auditing.

Objectives & Challenges

The Core Objective

Establishing a data-driven framework to transform massive synthetic patient logs into operational reliability.

The Business Challenge

Mitigating the rise in 30-day hospital readmissions by identifying high-risk profiles before discharge occurs.

The Technical Ecosystem



Python

Data cleaning, imputation of missing values, and feature engineering via Google Colab environments.



Microsoft SQL

Exploratory data analysis, demographic baselines, and bottleneck identification in SSMS.



Power BI

Interactive dashboarding and reporting for clinical intervention and high-level risk assessment.

Data Pre-processing

Refining the Raw Stream

Utilizing Python for rigorous data inspection and imputation. We handled complex null values and standardized data types across diverse medical categories. The output was a high-integrity cleaned dataset ready for high-performance SQL querying and secondary analysis.



| Exploratory Analysis (SQL)



Demographic Baselines: Segmenting patients by age, gender, and pre-existing conditions to understand core population traits.



Resource Utilization: Identifying bottlenecks in hospital capacity and specialized medical center efficiency.



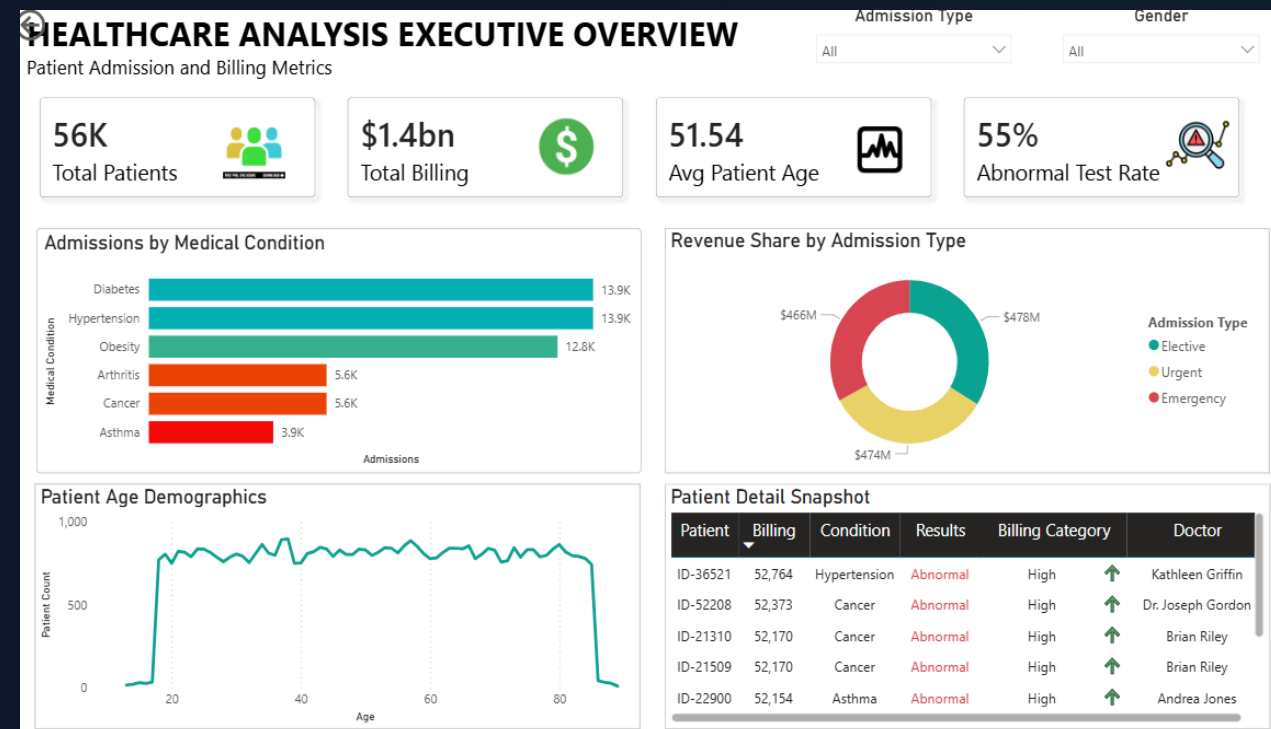
Anomaly Detection: Pinpointing high-impact outliers and potential data integrity issues within transaction logs.

Operational Visual Analytics

Power BI Dashboard

Delivered an interactive reporting ecosystem that pinpointed high-risk groups in real-time. This allows for proactive clinical interventions and significant long-term cost savings.

Key metrics focus on 30-day readmission risk factors and financial auditing accuracy across various billing categories.



Clinical Inferences

Chronic Conditions

Hypertension and Diabetes consistently appear as the primary drivers for recurring hospital admissions.

Action: Launch a Chronic Disease Management Program (CDMP) for remote monitoring.

Aging Demographic

A vast majority of admissions occur within the 50-75+ age bracket, increasing resource pressure.

Action: Invest in Geriatric Specialized Care and facility infrastructure upgrades.

| The Path to Risk Mitigation

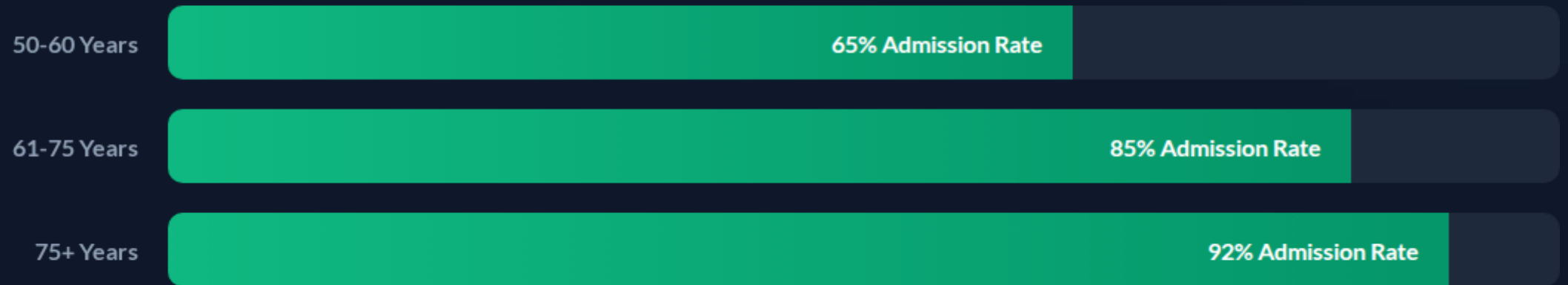
50%

Abnormal Test Rate

Automated Follow-up Systems

With test results frequently indicating high-risk status, the system must automate specialist consultation scheduling. Automating these touchpoints within 7 days of discharge directly targets and reduces the 30-day readmission challenge identified earlier.

Predictive Facility Efficiency



Predictive staffing models must be deployed to handle the volatility of the aging demographic, ensuring nurse-to-patient ratios are optimized for these high-need segments.

Efficiency & Impact Outlook



Scaling Operational Success

By integrating data integrity with operational reliability, we achieve a triple win: improved patient outcomes through clinical intervention, reduced costs via efficient staffing, and optimized financial returns through revenue auditing.

The dashboard acts as the central hub for this transition from reactive to proactive care.